

Extra Worked Examples NP

1. Wilcoxon Signed Rank Sum Test

- 1.1) Nine pairs of matched students are tracked through a one-semester course in mathematics. The first member of each pair is provided with free tutoring as needed. Otherwise, all students have the same instructor test, tests, and so on. Their class grades as percentages of the total number of possible points are given in the table below. Does the free tutoring significantly affect the classroom performance of students (assuming matches was done on variables known to be highly correlated with classroom performance)?

Pair	X	Y	Diff	Diff	Rank	Signed Rank
1	63	71				
2	74	74				
3	78	75				
4	98	90				
5	77	58				
6	85	89				
7	62	70				
8	83	69				
9	61	60				
10	64	62				
11.	82	81				

SOLUTION:

Hypothesis statements:

H0: There is no difference in classroom performance between the students who received free tutoring and those who did not. In other words, tutoring had no effect.

H1: There is a difference in classroom performance between the two groups, tutoring did have an effect.

Test statistic

- Calculate paired differences (table)
- Remove difference that are zero
- Record n (the number of non-zero paired differences) = 10 (one removed)
- Rank absolute difference

Pair	X	Y	Diff	Diff	Rank	Signed Rank
1	63	71	-8	8	7	-7
2	74	74	0	removed	removed	removed
3	78	75	3	3	4	4
4	98	90	8	8	7	7
5	77	58	19	19	10	10
6	85	89	-4	4	5	-5
7	62	70	-8	8	7	-7
8	83	69	14	14	9	9
9	61	60	1	1	1.5	1.5
10	64	62	2	2	3	3
11.	82	81	1	1	1.5	1.5

e. Calculate W (sum of signed ranks) = -7+4+7+10-5-7+9+1.5+3+1.5 = 17

Critical region

The number of paired differences is greater 10, so we use the normal approximation.

$$z = \frac{W - \mu_w}{\sigma_w} = \frac{17 - 0}{\sqrt{385}} = 0.8664 \approx 0.87$$

$$\sigma_w = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{10(10+1)(2(10)+1)}{6}} = \sqrt{385}$$

From table:

$$p\text{-value} = 1 - 0.8051 = 0.1949 * 2 = 0.3898 \text{ (two-sided test)}$$

Conclusion

Since the p-value of 0.3898 is greater than the significance level of 0.05, we do not reject the null hypothesis and conclude that free tutoring did not have an effect on classroom performance.

- 1.2) Eight male and eight female executives are matched as to age, education, duration of employment, type of job, and size of company with which they are employed. The salaries of these eight matched pairs are given below. Test the hypothesis that the female executives receive lower salaries, with a significance level of 5%.

Pair	Male	Female	Diff	Diff	Rank	Signed Rank
1	53,500	46,250				
2	37,200	40,000				
3	75,500	62,750				
4	45,900	44,000				
5	96,800	82,400				
6	39,100	44,500				
7	55,800	49,500				
8	84,200	76,200				
9	60,000	43,200				
10	54,000	41,000				
11	55,000	41,500				
12	74,300	75,000				

SOLUTION:

Hypothesis statements:

H0: There is no difference in salary between matched male and female executives, biological sex has no effect on executive salary.

H1: Biological sex does have an effect on the salary of executives, specifically male executives receive higher salaries than their female counterparts. (one-sided, upper tail test).

Test statistic

- a. Calculate paired differences (table)

- b. Remove difference that are zero
- c. Record n (the number of non-zero paired differences) = 12
- d. Rank absolute difference

Pair	Male	Female	Diff	Diff	Rank	Signed Rank
1	53,500	46,250	7250	7250	6	6
2	37,200	40,000	-2800	2800	3	-3
3	75,500	62,750	12750	12750	8	8
4	45,900	44,000	1900	1900	2	2
5	96,800	82,400	14400	14400	11	11
6	39,100	44,500	-5400	5400	4	-4
7	55,800	49,500	6300	6300	5	5
8	84,200	76,200	8000	8000	7	7
9	60,000	43,200	16800	16800	12	12
10	54,000	41,000	13000	13000	9	9
11	55,000	41,500	13500	13500	10	10
12	74,300	75,000	-700	700	1	-1

- e. Calculate W (sum of signed ranks) = $6 - 3 + 8 + 2 + 11 - 4 + 5 + 7 + 12 + 9 + 10 - 1 = 62$

Critical region

The number of paired differences is greater 10, so we use the normal approximation.

$$z = \frac{W - \mu_w}{\sigma_w} = \frac{62 - 0}{5\sqrt{26}} = 2.4318 \dots \approx 2.43$$

$$\sigma_w = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{12(12+1)(2(12)+1)}{6}} = 5\sqrt{26}$$

From table:

$$p\text{-value} = 1 - 0.9925 = 0.0075$$

Conclusion

Since the p-value of 0.0075 is smaller than the significance level of 0.05, we reject the null hypothesis and conclude that male executives do receive higher salaries than their female counterparts.

2. Mann Whitney U-test

2.1) The nitrous oxide emissions, measured in parts per million (ppm), for samples from two different engine types (1 and 2) are summarized below. Test the hypothesis that the nitrous oxide emissions from the first engine are greater than those from the second engine.

Type 1	1.41	1.35	1.3	1.2	1.05
Type 2	1.28	1.25	1.15	1.1	

SOLUTION

Hypothesis statements

H0: There is no difference in the distribution of nitrous oxide emissions between the first and second engine type.

H1: The first engine has greater nitrous oxide emissions than the second engine type. The distribution of nitrous oxide emissions for the first engine type has a greater median than the nitrous oxide emission distribution of the second engine type.

Test statistic

- Combine two samples ($n_1 = 5$ and $n_2 = 4$)
- Rank all observations
- Calculate sum of ranks within each sample

1.41	1.35	1.3	1.2	1.05	1.28	1.25	1.15	1.1
9	8	7	4	1	6	5	3	2

$$T_1 = 29$$

$$T_2 = 16$$

Check:

$$T_1 + T_2 = 45$$

$$\frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} = \frac{(5 + 4)(5 + 4 + 1)}{2} = 45$$

Critical region

N_1 (5) and n_2 (4) are both smaller than 10, so we use table of exact probabilities and not approximated p-value. Reject if T less than or equal to T_L or greater than or equal to T_U .

$$T_L = 17$$

$$T_U = n_1(n_1 + n_2 + 1) - T_L = 5(5 + 4 + 1) - 17 = 58$$

Conclusion

Since $17 < T_1 = 29 < 58$, we do not reject the null hypothesis and conclude that there is no significant difference in nitrous oxide emissions between the two engine types.

2.2) Eight left-handed and 10 right-handed third grade students were selected and coached on several new mathematical concepts until all were mastered by each student. Eight days later they were given a mastery test. The results are given below. Using a MWU test with alpha equal to .05, test for difference in the ability of left- and right-handed students to retain mathematical concepts.

Right	64	78	82	84	88	90	91	91		
Left	66	68	70	75	81	81	84	86	87	91

SOLUTION

Hypothesis statements

H0: There is no difference in the distribution of results between right-handed and left-handed students with regards to the mastery tests.

H1: There is a difference in the mastery results between left- and right-handed student.

Test statistic

- Combine two samples ($n_1 = 8$ and $n_2 = 10$)
- Rank all observations
- Calculate sum of ranks within each sample

64	78	82	84	88	90	91	91	66	68	70	75	81	81	84	86	87	91
1	6	9	10.5	14	15	17	17	2	3	4	5	7.5	7.5	10.5	12	13	16

T1 = 89.5

T2 = 64.5

Check:

$$T1 + T2 = 154$$

$$\frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} = \frac{(8 + 10)(8 + 10 + 1)}{2} = 154$$

Critical region

N2 is greater than or equal to 10, so we use the normal approximation approach.

$$Z = \frac{T1 - \mu_T}{\sigma_T} = \frac{89.5 - 76}{\sigma_T} = 1.1995 \dots \approx 1.20$$

$$\mu_t = \frac{n_1(n_1 + n_2 + 1)}{2} = \frac{8(8 + 10 + 1)}{2} = 76$$

$$\sigma_T = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{(8)(10)(8 + 10 + 1)}{12}} = \frac{2\sqrt{285}}{3}$$

Conclusion

Critical value = 1.65

p-value from lecture table = 1 - 0.8849 = 0.1151

Since the p-value is greater than the significance level of 5% (can also say since the test statistic is smaller than the critical value), we do not reject the null hypothesis and conclude that there is no difference between right-handed and left-handed mastery results.

3. Kruskal-Wallis

- 3.1) Random samples of graduates from the engineering, psychology and sociology departments of a major university were asked to rate on a scale of 0 to 100 their satisfaction with their degree. Using a 5% significance level and the data below, test the hypothesis that the degree of satisfaction is the same for all 3 departments.

Engineering	Psychology	Sociology
84	55	76
81	57	72
75	64	61
65	73	60
52	80	58
50	84	52
50	93	

Hypothesis statements

H₀: There is no difference in the ratings given by students between the three departments.

H₁: There is a difference in the ratings given by students between the three departments.

Test statistic

k = 3

n₁ = n₂ = 7; n₃ = 6

n_T = 20

- a. Rank combined sample

Engineering	Psychology	Sociology
84 18.5	55 5	76 15
81 17	57 6	72 12
75 14	64 10	61 9
65 11	73 13	60 8
52 3.5	80 16	58 7
50 1.5	84 18.5	52 3.5
50 1.5	93 20	
T ₁ = 67	T ₂ = 88.5	T ₃ = 54.5

- b. Check:

$$T_1 + T_2 + T_3 = 210$$

$$\frac{n_T(n_T + 1)}{2} = \frac{20(20 + 1)}{2} = 210$$

$$H = \left[\frac{12}{n_T(n_T + 1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1) = \left[\frac{12}{20(20 + 1)} \left(\frac{67^2}{7} + \frac{88.5^2}{7} + \frac{54.4^2}{6} \right) \right] - 3(20 + 1) = 1.43$$

Critical region

$$H \sim \chi_{k-1}^2$$

Where k = 3

$$\chi_{k-1}^2 = \chi_{2,0.05}^2 = 5.991$$

Conclusion

Since the test statistic or 1.43 is smaller than the chi-squared critical value of 5.991, we do not reject the null hypothesis at the 5% significance level and conclude that there is no difference in ratings given by students between the three departments.

p-value is greater than 0.1 (from table)

- 3.2) The sugar content is measured in randomly selected samples of each of four brands of certain type of breakfast cereal. The measurements (grams of sugar per pound of cereal) are given below. State and test a hypothesis using a 1% significance level.

Cereal A	Cereal B	Cereal C	Cereal D
85.4	86.7	85.1	85.6
86.5	87.1	86.9	85.6
86.5	87.2	88	85.9
86.8	87.4	88	86.1
87	87.4	88.3	86.2
87.1	87.5	88.4	86.4
	87.8	88.6	86.4
	88.1	88.7	86.4
		88.9	86.6
			86.9
			87.9

Hypothesis statements

H₀: There is no difference in the sugar content between the four cereal brands.

H₁: There is a difference in the sugar content between the four cereal brands.

Test statistic

k = 4

n₁ = 6, n₂ = 8, n₃ = 9, n₄ = 11

n_T = 34

- a. Rank combined sample

Cereal A	Cereal B	Cereal C	Cereal D
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85.4 2	86.7 14	85.1 1	85.6 3.5
86.5 11.5	87.1 19.5	86.9 16.5	85.6 3.5
86.5 11.5	87.2 21	88 27.5	85.9 5
86.8 15	87.4 22.5	88 27.5	86.1 6
87 18	87.4 22.5	88.3 30	86.2 7
87.1 19.5	87.5 24	88.4 31	86.4 9
	87.8 25	88.6 32	86.4 9
	88.1 29	88.7 33	86.4 9
		88.9 34	86.6 13
			86.9 16.5
			87.9 26
T1 = 77.5	T2 = 177.5	T3 = 232.5	T4 = 107.5

b. Check:

$$T1 + T2 + T3 + T4 = 595$$

$$\frac{n_T(n_T + 1)}{2} = \frac{34(34 + 1)}{2} = 595$$

$$H = \left[\frac{12}{n_T(n_T + 1)} \sum_{i=1}^k \frac{T_i^2}{n_i} \right] - 3(n_T + 1)$$

$$= \left[\frac{12}{34(34 + 1)} \left(\frac{77.5^2}{7} + \frac{177.5^2}{8} + \frac{232.5^2}{9} + \frac{107.5^2}{11} \right) \right] - 3(34 + 1) = 14.52721$$

Critical region

$$H \sim \chi_{k-1}^2$$

Where k = 4

$$\chi_{k-1}^2 = \chi_{3,0.01}^2 = 11.345$$

Conclusion

Since the test statistic of 14.53 is greater than the chi-squared critical value of 11.35, we reject the null hypothesis at the 1% significance level and conclude that there is a difference in sugar content between the cereal brands.

p-value is smaller than 0.005 (from table)